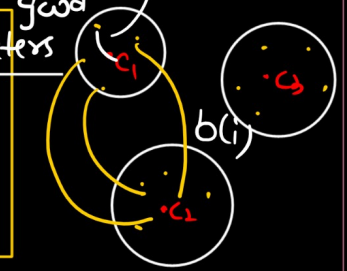


Evaluation Metric

$b(i) \gg a(i) \rightarrow$ to have good clusters

$$\text{Silhouette score} \Rightarrow \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

score $\rightarrow -1$ to $+1$



Intra-cluster distance (a_i) : This is the average distance between the data point and all other points in the same cluster. A smaller value indicates the point is closely aligned with its cluster.

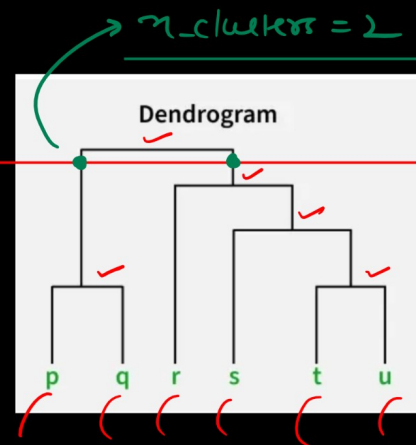
Nearest-cluster distance (b_i) : This is the average distance between the data point and all points in the nearest neighbouring cluster (the next best alternative). A larger value means the point is well-separated from other clusters.

Hierarchical Clustering

Single Linkage

Average Linkage

Ward Linkage



small

variation in

the value of WCSS

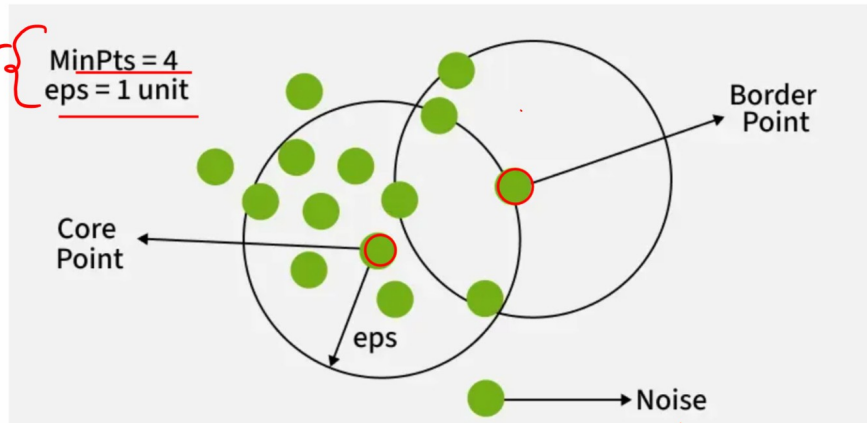
DBSCAN $\longrightarrow$ minpts eps

DBSCAN works by categorizing data points into three types:

1. Core points which have a sufficient number of neighbors within a specified radius (epsilon)
2. Border points which are near core points but lack enough neighbors to be core points themselves
3. Noise points which do not belong to any cluster.

By iteratively expanding clusters from core points and connecting density-reachable points, DBSCAN forms clusters without relying on rigid assumptions about their shape or size.

hyperparameters
tuning



Outlier